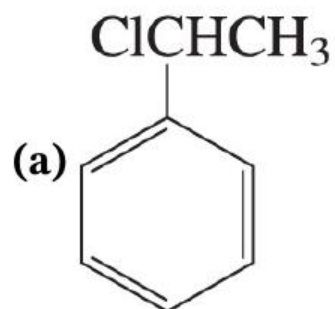
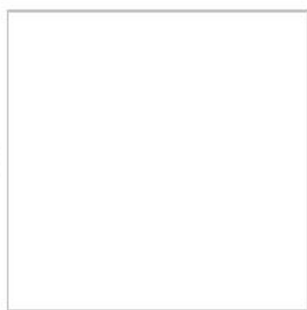


Solutions to Problems

37.

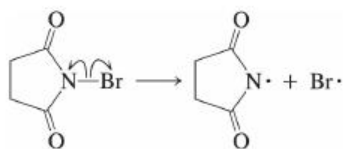


(b)

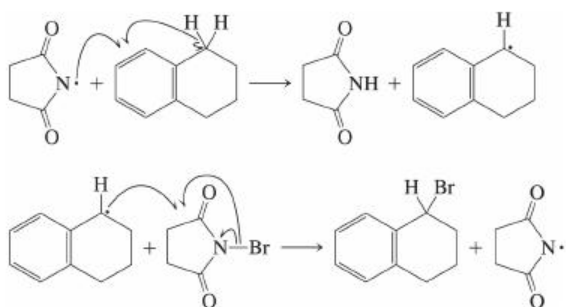


38. Radical chemistry: initiation, propagation, and termination (which we will ignore).

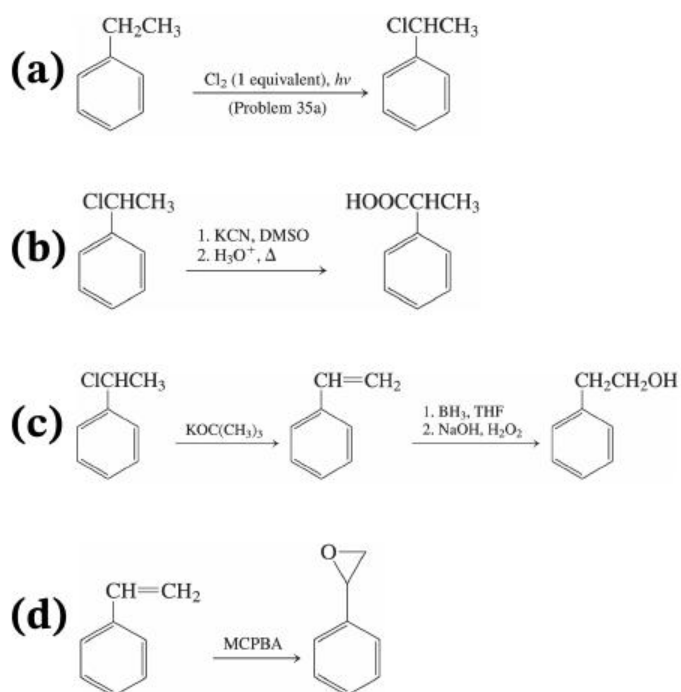
INITIATION:

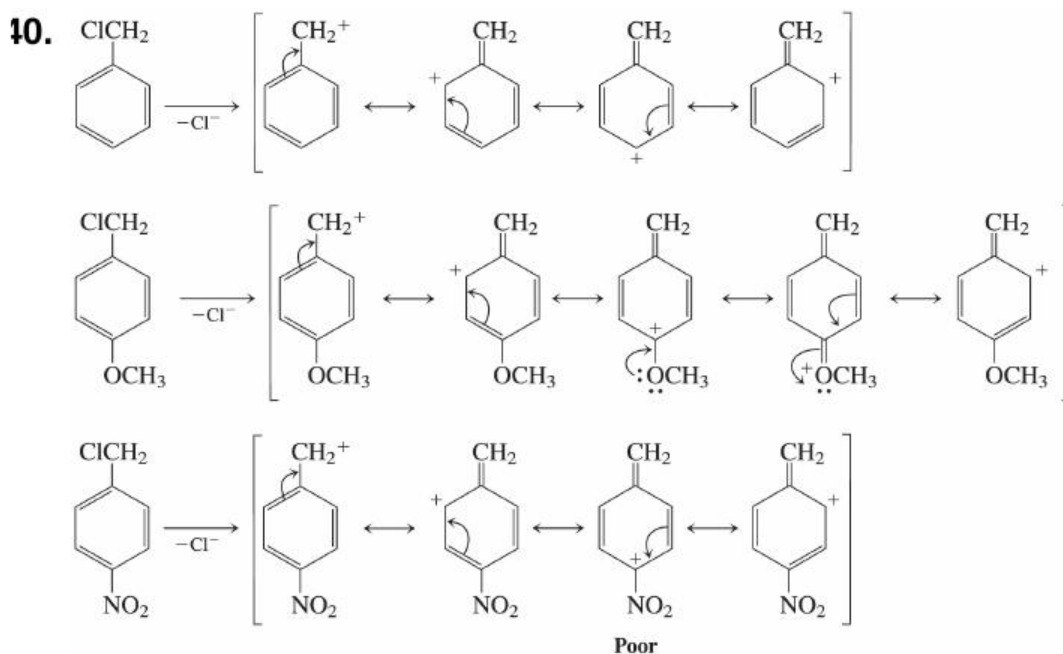


PROPAGATION:

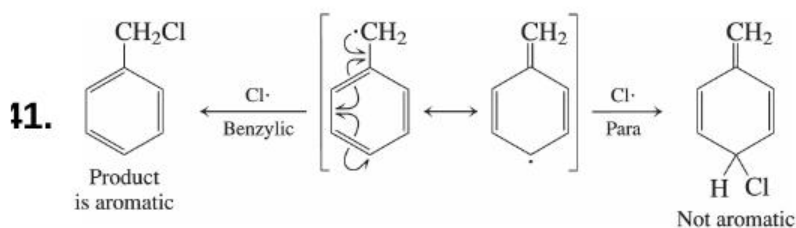


39. Several reactions derive from fundamental material presented in earlier chapters.



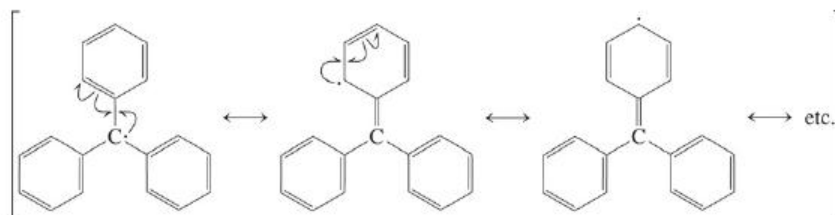


The cation derived from chloromethylbenzene (benzyl chloride, top) is stabilized by four resonance contributing forms. The cation from 1-(chloromethyl)-4-methoxybenzene (4-methoxybenzyl chloride, middle) is more stable because of the added contributor in which a lone pair on oxygen is delocalized into the ring (second Lewis structure from the right). The cation from 1-(chloromethyl)-4-nitrobenzene (4-nitrobenzyl chloride, bottom) is least stable because one of its resonance forms, the one with the + charge next to the electron-withdrawing nitro group, is poor.

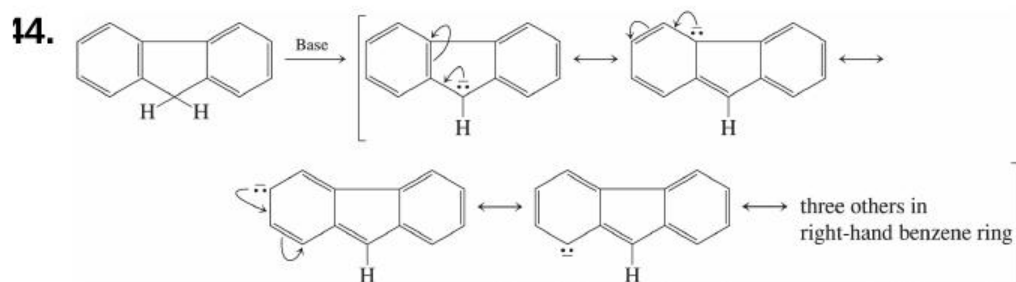
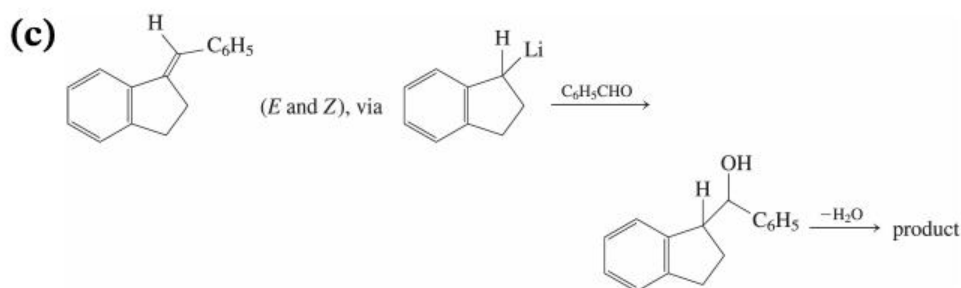
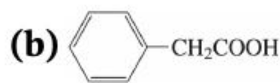


12. In each case numerous resonance forms can be drawn. Both of these species are resonance hybrids strongly stabilized by delocalization of the charge (in the cation) or the odd electron (in the radical). Three of the resonance forms for the radical are shown below. How many

more can you draw?

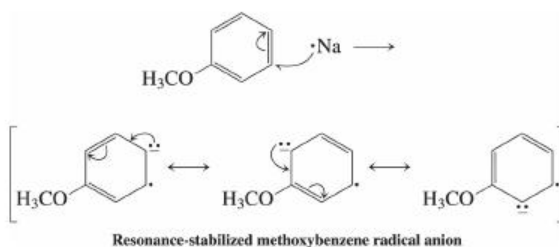


13. (a) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{---}\text{C}_6\text{H}_4\text{---CH}_2\text{OH}$ Benzylic position is most reactive in nucleophilic substitution.

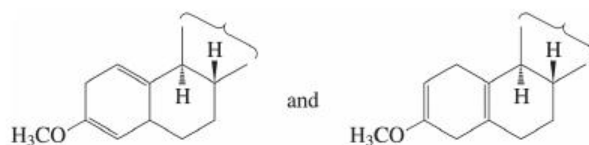


Seven resonance forms make this carbanion especially stable. It is also aromatic, having 14π electrons in an unbroken loop of p orbitals.

15. Birch reduction introduces one electron at a time into the benzene ring. Given the presence of an electron-donating methoxy group, the preference would be to introduce the first electron so as to place a negative charge at a ring position *not* occupied by that group:



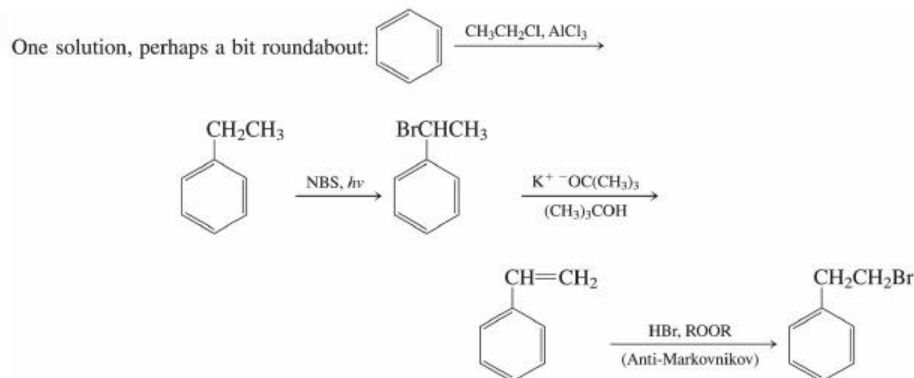
As mentioned in [Section 22-2](#) of the text, the tendency is for protonation to occur at the negative site in the middle resonance form. Then, after addition of a second electron and protonation, the product is the 1,4-diene, with the methoxy group on one of the double bonds. Depending on which formal benzene resonance form one uses as the starting structure, two possible products may be written for the actual molecule in the problem:



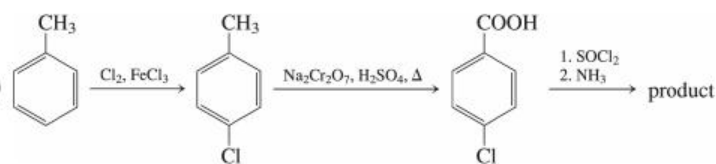
The NMR^{NMR} spectrum of the actual product shows only one alkene hydrogen signal, which is consistent with only the structure on the right. Its preferential formation may be explained by the presence of a more stable tetrasubstituted double bond.

16.

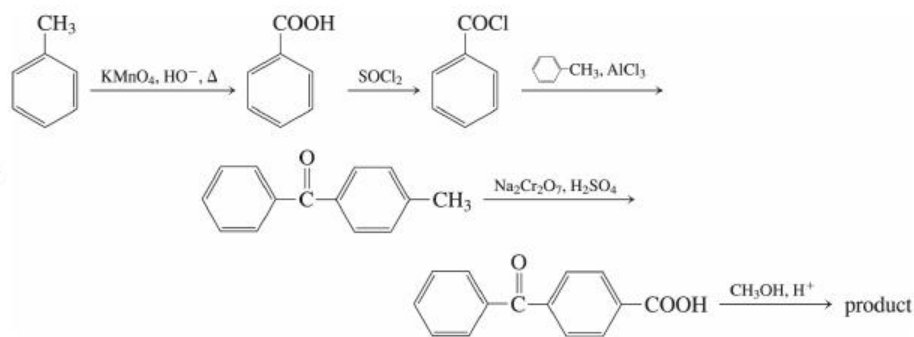
(a)



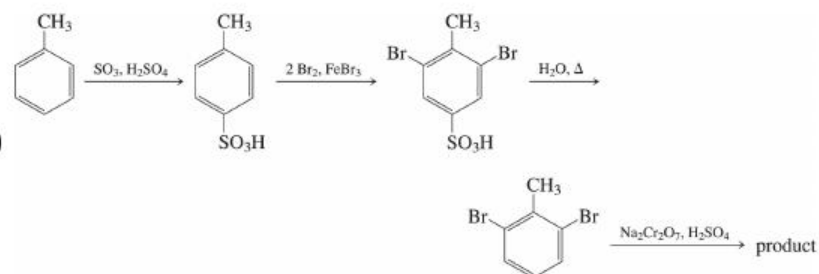
(b)



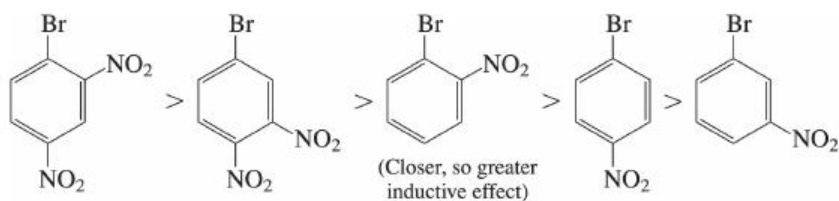
(c)



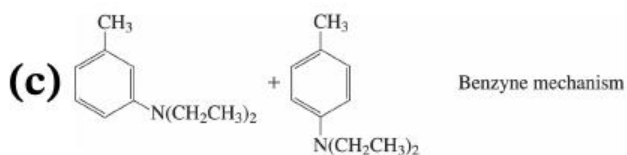
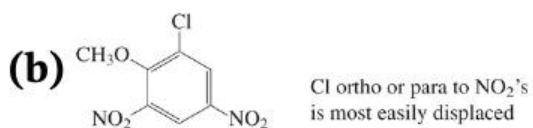
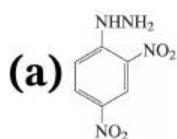
(d)



17. Most reactive ones have NO_2NO_2 groups ortho or para to the leaving group. So,



- 18.** The presence of strong electron-withdrawing groups such as NO_2 ortho or para to a potential leaving group favors nucleophilic aromatic substitution by the *addition-elimination* mechanism. Without such groups in these positions, the *benzyne* mechanism is favored.



19.

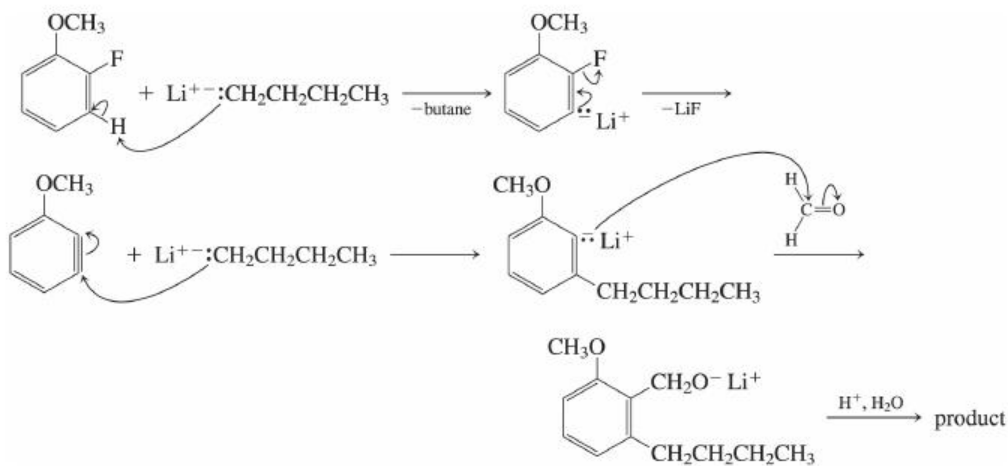
(a) 1. CH_3COCl , CH_3COCl (forms amide, reduces ring activation so that bromination in the next step can be controlled to attach just one Br to the ring instead of three), 2. Br_2 , CHCl_3 , Br_2 , CHCl_3 , 3. KOH , H_2O , Δ
 KOH , H_2O , Δ (hydrolyzes amide back to aniline);

(b) 1. $\text{CF}_3\text{CO}_3\text{H}$, CH_2Cl_2 , $\text{CF}_3\text{CO}_3\text{H}$, CH_2Cl_2 , 2. Cl_2 , FeCl_3 ;
 Cl_2 , FeCl_3 ;

(c) KCN , KCN (nucleophilic aromatic substitution);

(d) H^+ , H_2O , Δ , H^+ , H_2O , Δ

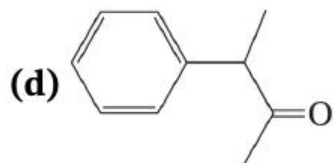
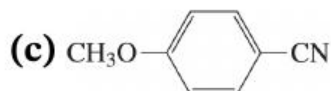
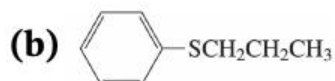
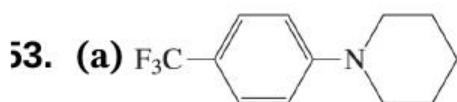
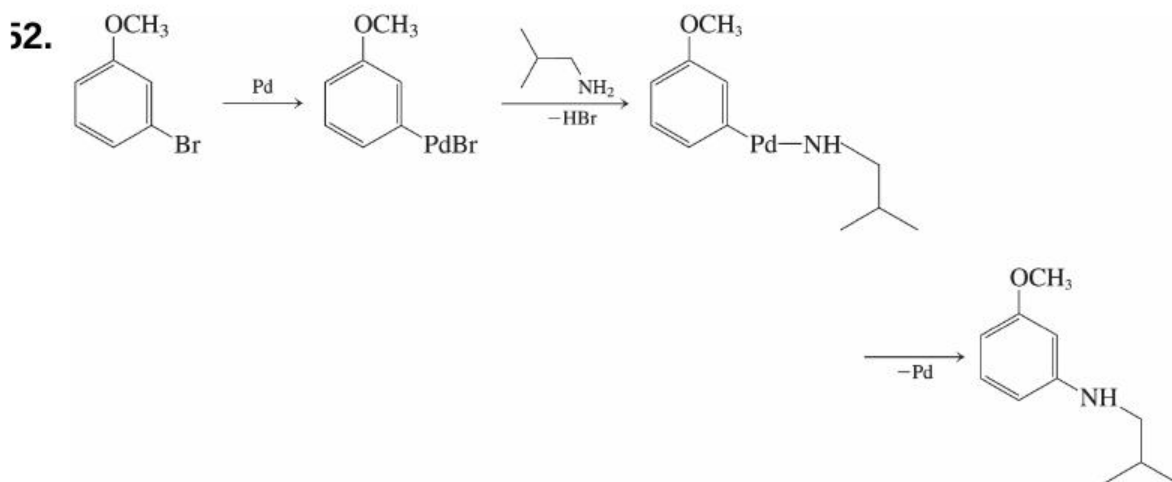
50.



Strongly basic butyllithium removes HF in the first steps, generating benzyne. A second mole of butyllithium adds to benzyne as a nucleophile. The reason for the direction of addition is explained in the answer to Exercise 22-14.

51. In nucleophilic aromatic substitution reactions that proceed by the addition-elimination mechanism, the *addition* step is rate-limiting.

Fluorine is the most electronegative and, therefore, the most electron-withdrawing of the halogens. Therefore, the presence of F lowers the transition-state energy for nucleophilic addition the most of all the halogens, and stabilizes the resulting anion by an inductive effect to the greatest degree. While it is true that $\text{F}-\text{F}^-$ is by far the worst leaving group among halide ions, expulsion of the leaving groups takes place *after* the rate-determining addition and is fast because it benefits from reestablishment of the aromaticity of the benzene ring. In [Chapter 25](#) we will see even worse leaving groups than this expelled to generate aromatic rings.



54. (a) $\text{Ph}_3\text{P}=\text{CH}_2$ (Wittig);

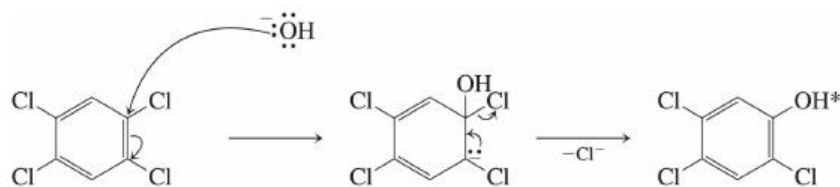
(b) $\text{CH}_3\text{COCl}, \text{NaOH}, \text{H}_2\text{O}$;



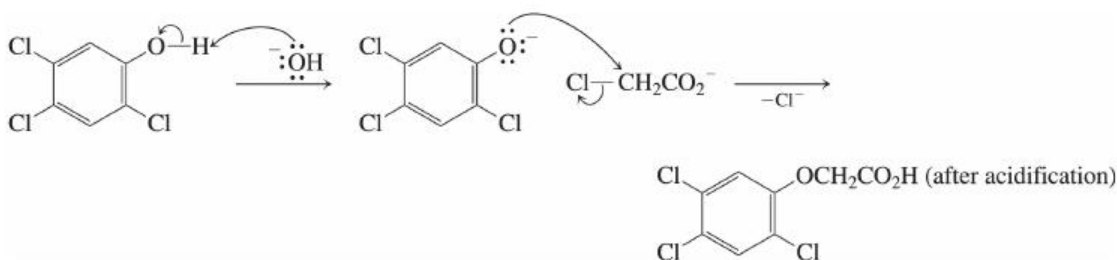
(c) $\text{Br}-\text{C}_6\text{H}_4-\text{OCOCH}_3$, $\text{Pd}(\text{OCOCH}_3)_2$, R_3P , 100°C (Heck)

(d) $\text{NaOH}, \text{H}_2\text{O}$.

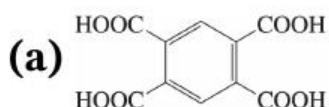
55. The first step introduces oxygen into the ring. Nucleophilic aromatic substitution is the most reasonable method. The resulting phenol must be used as a nucleophile itself to attack an appropriate substrate in order to give the final product.



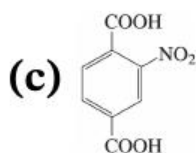
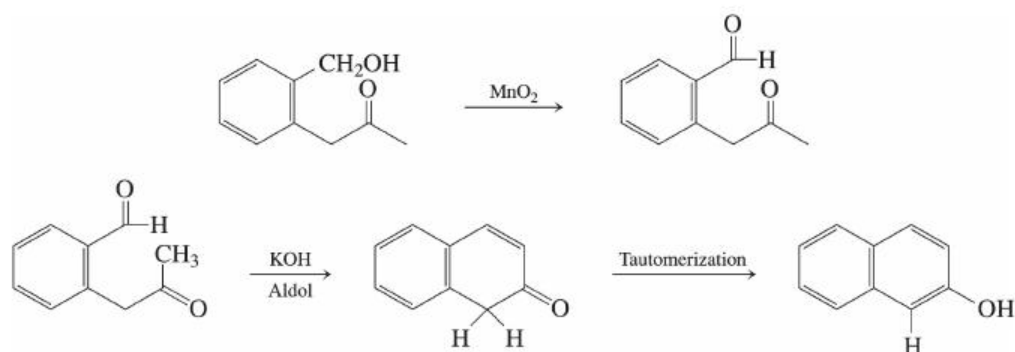
*Acidification is necessary because this phenolic hydroxy group will be deprotonated at the pH used for the initial nucleophilic aromatic substitution.



56.



- (b) The first step oxidizes the benzylic alcohol to an aldehyde. The molecule is then capable of an intramolecular aldol condensation, forming a second six-membered ring that contains an α,β -unsaturated ketone. This ketone, in turn, is unusual in that it is a cyclohexadienone, the ketone tautomer of a more stable phenol. Using equations, we have

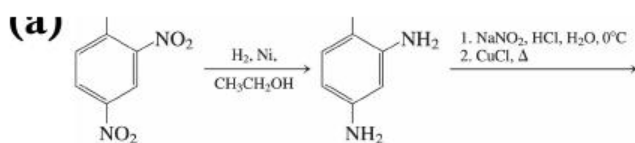


57.

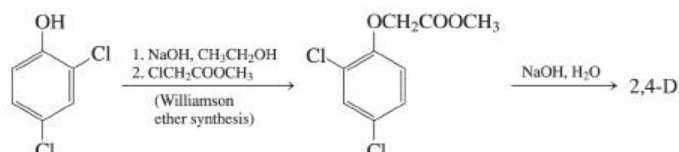
- (c) (The SO_3H group is very acidic) > (b) > (e) > (f) > (d) > (a).

(The SO_3H group is very acidic) > (b) > (e) > (f) > (d) > (a).

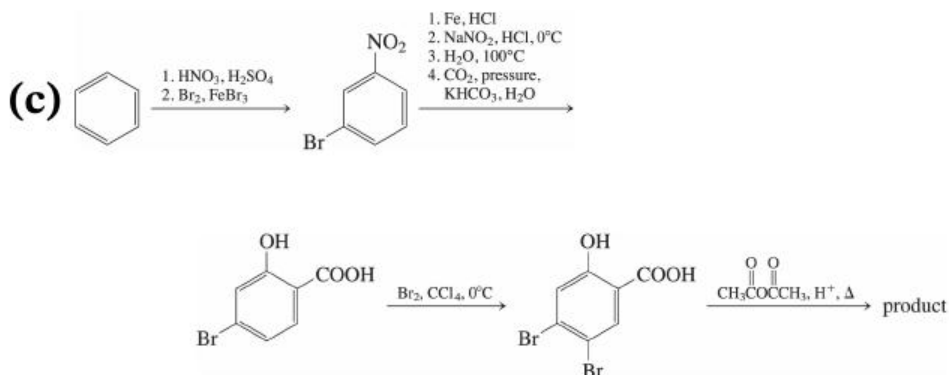
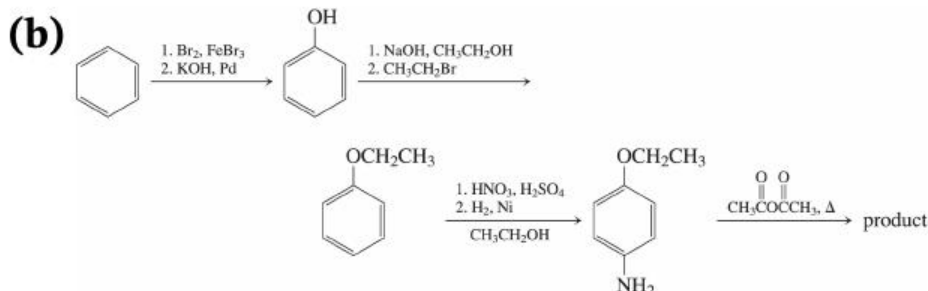
Carboxylic acids are more acidic than most phenols, and electron-withdrawing groups increase the acidity of phenols.



[From Problem 58(d)]



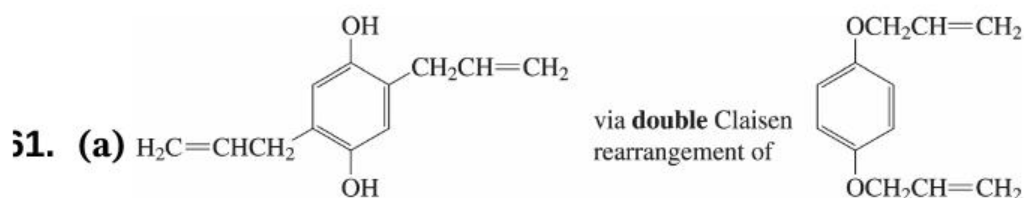
Can 2,4-dichlorophenol be prepared by direct dichlorination of phenol? Yes, but the two successive chlorination reactions are not ideally regioselective: Monochlorination of phenol gives both 2- and 4-chlorophenol, and the second chlorination of that mixture results in both the 2,4- and 2,6-disubstituted products in comparable amounts.



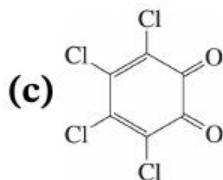
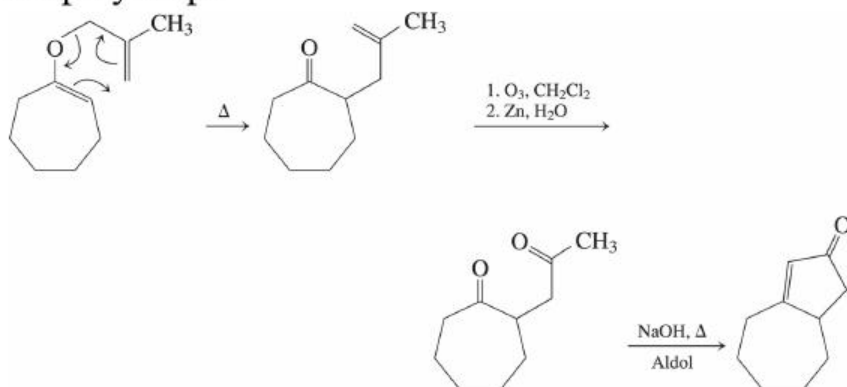
(d) Begin with Friedel-Crafts alkylation of benzene with 2-chloropropane and AlCl_3 . Then take the

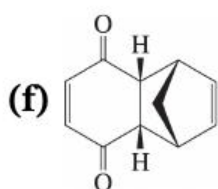
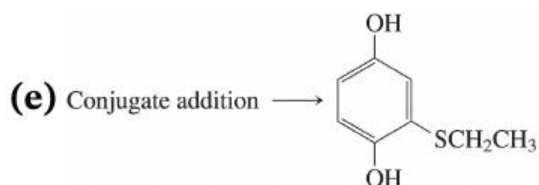
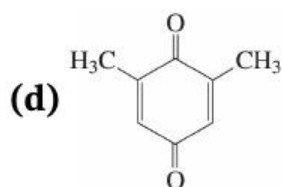
2-chloropropane and HCl . Then take the product (isopropylbenzene) through the steps for the synthesis of 1,3-benzenediol as shown in the answer to [Problem 58\(c\)](#) above.

30. (a) 5-Bromo-2-chlorophenol
 (b) 4-(hydroxymethyl)phenol
 (c) 2,4-dihydroxybenzenesulfonic acid
 (d) 2-phenoxyphenol
 (e) 2-methylthio-2,5-cyclohexadiene-1,4-dione
 (f) 2,6-dimethoxy-3,5-dimethylcyclohexa-2,5-diene-1,4-dione

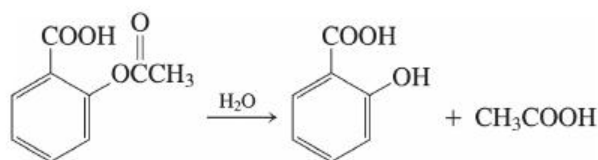


(b) Step-by-step:





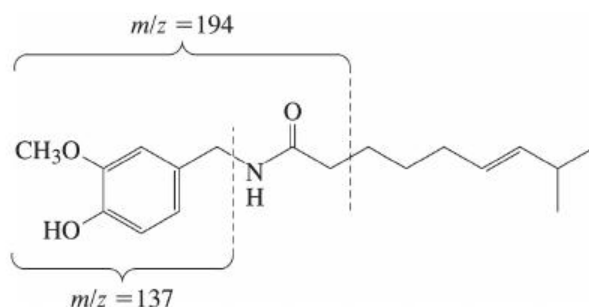
32. Aspirin is a *phenyl* ester. Phenyl esters are considerably more susceptible to hydrolysis than ordinary esters are, for two reasons: Delocalization of a lone electron pair from the phenol oxygen into the carbonyl group (ester resonance, see [Section 20-1](#)) is diminished, because this lone pair is also in resonance with the benzene ring. The result is a greater δ^+ on the carbonyl carbon, which facilitates nucleophilic attack. The electron-withdrawing COOH group accentuates this effect. The second reason is thermodynamic: The equilibrium $\text{phenol} + \text{carboxylic acid} \rightleftharpoons \text{phenyl ester}$ favors the starting material ([Section 22-5](#)). So, aqueous solutions of aspirin hydrolyze rather rapidly at room temperature to give salicylic acid and acetic acid.



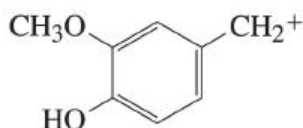
Tylenol, on the other hand, is an amide and much more resistant to nucleophilic attack. Hydrolysis would require extended heating, and

strong acid or base is required. [Indeed, amides derived from aniline, such as *NN*-acetylaniline (acetanilide), are commonly purified by recrystallization from neutral boiling water, further illustrating their stability.]

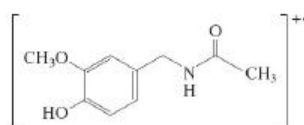
33. Without an exact mass measurement the molecular ion at $m/z=305$ isn't much use for formula determination (there are far too many possibilities). However, working back from the actual structure we can see where the major fragmentation peaks come from. In the structure below we've inserted the masses of several portions of the molecule, obtained by just adding the integral masses of the atoms (H, 1; C, 12; N, 14; and O, 16).



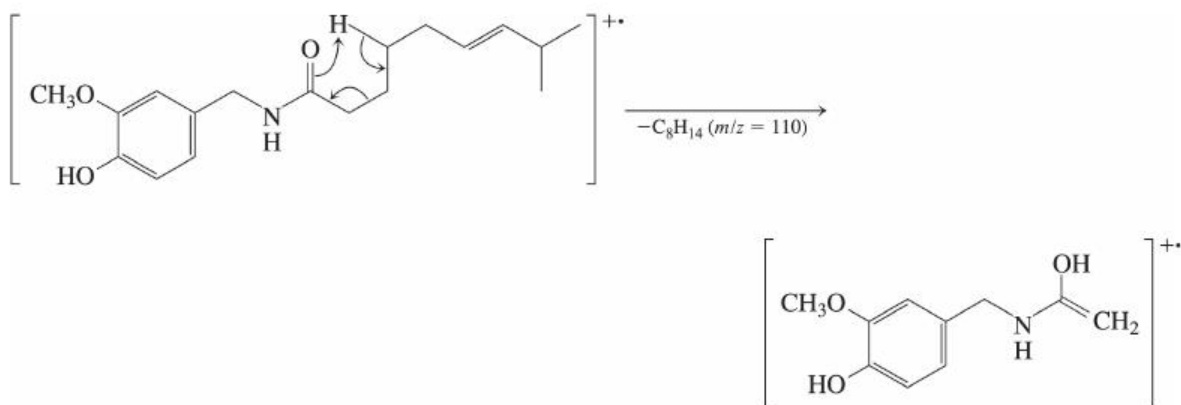
The base peak at $m/z=137$ is recognizable right away: Cleavage at the benzylic carbon gives the resonance-stabilized cation



from which loss of a methyl group ($m/z=15$) gives the $m/z=122$ peak. Meanwhile the “tricky” peak at $m/z=195$ is related to the $m/z=194$ fragment identified above; in fact, it is that piece plus a hydrogen. That is,



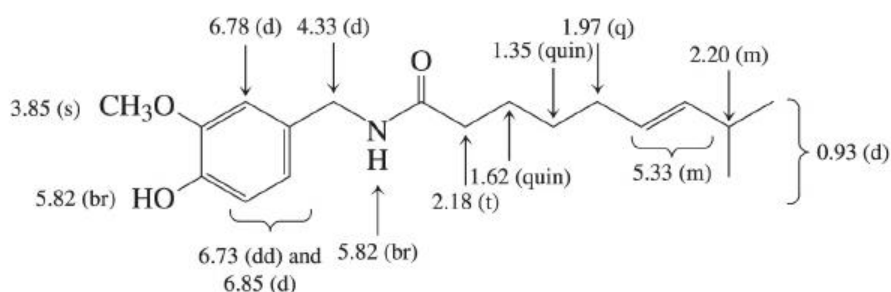
Where did that come from? Recall the McLafferty fragmentation of carbonyl compounds: It results in cleavage between the α and β carbons according to the following process:



which is the enol isomer of the methyl ketone structure shown on the preceding page.

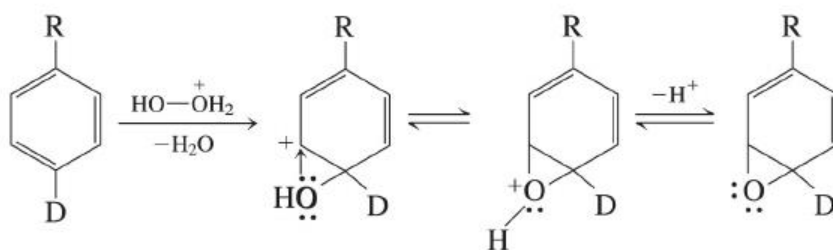
The IR data is fairly straightforward to interpret. The band at 972 cm^{-1} derives from a bending motion of the trans alkene double bond, that at 1660 cm^{-1} from the amide $\text{C}=\text{O}$ stretch, 3016 cm^{-1} from the alkene and arene $\text{C}-\text{H}$ stretches, and 3445 and 3541 cm^{-1} from the amide $\text{N}-\text{H}$ and phenol $\text{O}-\text{H}$ stretches.

Finally, the NMR spectrum contains a wealth of information with virtually every individual group of hydrogens clearly resolved. The assignments are made easier if you pay attention to the integrations and the splittings, which conform to the $N+1$ rule. Here they are (br=broad singlet; m=multiplet; quin=quintet):
(br = broad singlet; m = multiplet; quin = quintet) :

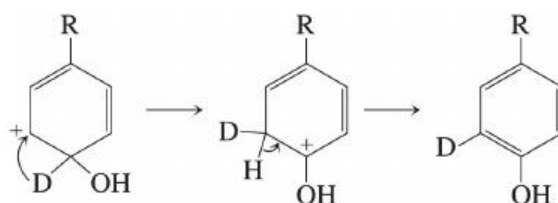


The OH and NH hydrogens are indistinguishable to the NMR machine in this molecule and combine to give a broad signal at δ 5.82 ppm. Unusually, the NH hydrogen splits the neighboring benzylic CH_2 group, whose signal at δ 4.33 ppm is a doublet (this assignment must be made by process of elimination—no other CH_2 group in the molecule can give a doublet in the NMR).

34. (a) Benzene rings containing only alkyl substituents are not very susceptible to attack by nucleophiles or radicals. Electrophilic attack is the most reasonable pathway.
- (b) Formation of the oxacyclopropane probably involves electrophilic attack of a reagent like H_2O_2 (derived from O_2) on the benzene ring, catalyzed by acid.

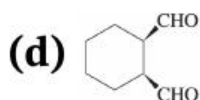
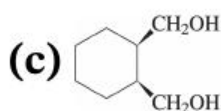
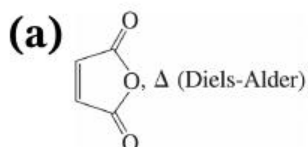


The final phenol probably arises from reversal of the last two steps. The oxacyclopropane ring can always close again, but eventually the carbocation reacts via an alternative pathway, involving migration to give the rearranged aromatic product.

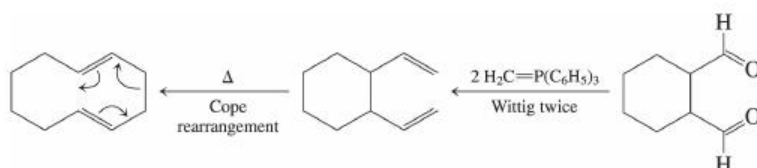


$D+D^+$ can also be lost in the final step, giving a phenol lacking deuterium.

35.



Now what? Work **backward** from the product.

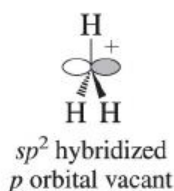


36. The normal “backside displacement” geometry associated with an SN_2 transition state is impossible to achieve at a carbon atom in a benzene ring. There is, in fact, no evidence to support the occurrence of the one-step SN_2 displacement mechanism at a benzene carbon.

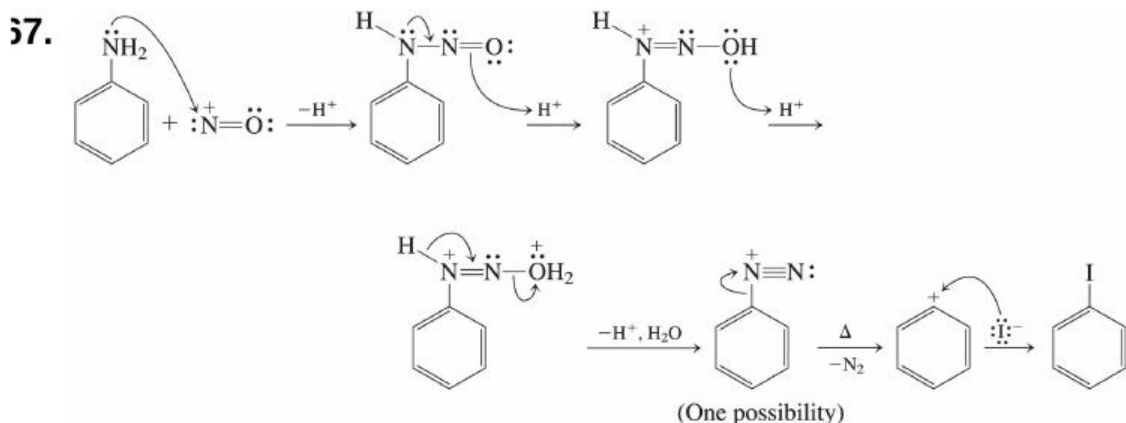
The SN_1 pathway requires formation of a carbocation. We saw

earlier ([Section 13-9](#)) that alkenyl halides are unreactive toward displacement reactions. The $\text{S}_{\text{N}}2$ pathway is poor for the same geometric reason we stated above for benzene compounds. In addition, alkenyl cations are high-energy species because their positive charge derives from an empty sp^2 orbital. The same issue arises in the case of benzene: The phenyl cation is a high-energy species and forms only with difficulty, because it contains a positively charged carbon atom with an empty sp^2 orbital.

Why is an empty sp^2 orbital bad? It contains partial s orbital character, and s orbitals are closer to the nucleus than p orbitals, which have a node at the positively charged atomic nucleus. Loss of an electron from a hybrid orbital with s character will therefore be more difficult than from a p orbital. Recall that all carbocations we have seen since [Chapter 7](#) have unoccupied p orbitals:



As a result of the difficulty of formation of phenyl cations, Sandmeyer-type substitutions of nitrogen in arenediazonium salts by Cl , Br , and CN follow an alternative, copper-catalyzed radical pathway. However, reaction to displace N_2 by either water or iodide does not require copper catalysis. Such observations support the possibility that an $\text{S}_{\text{N}}1$ mechanism is involved and that a phenyl cation is an intermediate. In the formation of phenols by reaction of arenediazonium salts with hot water, heat may facilitate loss of N_2 to give the cation; water as solvent would attack this species rapidly. Iodide, if present, is a much better nucleophile and would outcompete water for reaction with the cation. The solution to [problem 67](#), below, explores this possibility in more detail.



It is an open question whether a phenyl cation is actually involved in reactions such as this. One alternative is addition–elimination ([Section 22-4](#)), but this option is not supported by kinetic evidence, which indicates that the displacement process is not bimolecular but unimolecular. Another possibility is a radical process, initiated by reduction of the phenyldiazonium cation to a radical by the iodide ion, which is a good reducing agent. This mechanism is more likely because it has a unimolecular rate-determining propagation step, loss of N_2 from the phenyldiazo radical to give a phenyl radical, a much easier species to form than the phenyl cation.

Although the exact mechanism for iodobenzene formation from the diazonium cation is unclear, the phenyl cation is not an *impossible* intermediate. It is still considered to be the best candidate for the intermediate in phenol formation from arenediazonium salts and hot water ([Section 22-4](#)). In addition, it is known that diazonium cations exchange their bonded N_2 with gaseous N_2 (detected in isotope studies), direct evidence that reversible dissociation of N_2 to leave behind a phenyl cation can take place. This system is an example of how a seemingly simple structure can exhibit a lot of complexity in its behavior.

38. Compounds (a) through (d) require nitrosation (also called **diazotization**, using $\text{NaNO}_2, \text{HCl}$, 0°C to 5°C) to give the 3-methylbenzenediazonium salt, followed by treatment as appropriate

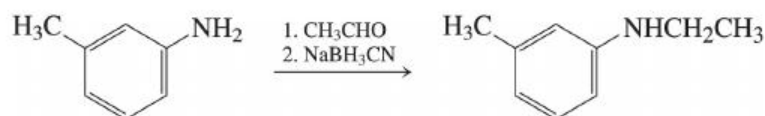
for the final desired substituent:

(a) $\text{H}_3\text{PO}_2/\text{H}_3\text{PO}_2$ (replaces $\text{N}_2^+/\text{N}_2^+$ with H);

(b) $\text{CuBr}, \text{CuBr}, 100^\circ\text{C}/100^\circ\text{C}$ (for BrBr);

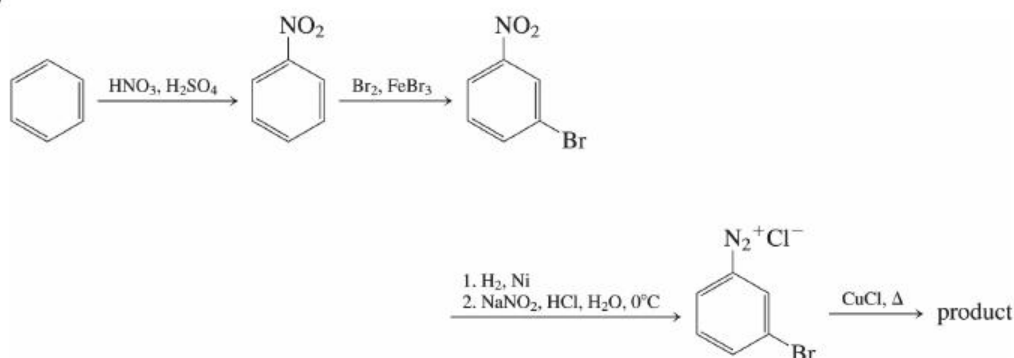
(c) $\text{H}_2\text{O}, \text{H}_2\text{O}$, heat (for OHOH);

(d) $\text{CuCN}, \text{KCN}, \text{CuCN}, \text{KCN}, 50^\circ\text{C}/50^\circ\text{C}$ (for CNCN). For (e) we are placing a substituent *on the nitrogen* rather than replacing the nitrogen with something else. Specifically, the problem asks for the conversion of a primary amine into a secondary amine, which is best done by reductive amination ([Section 21-6](#)):

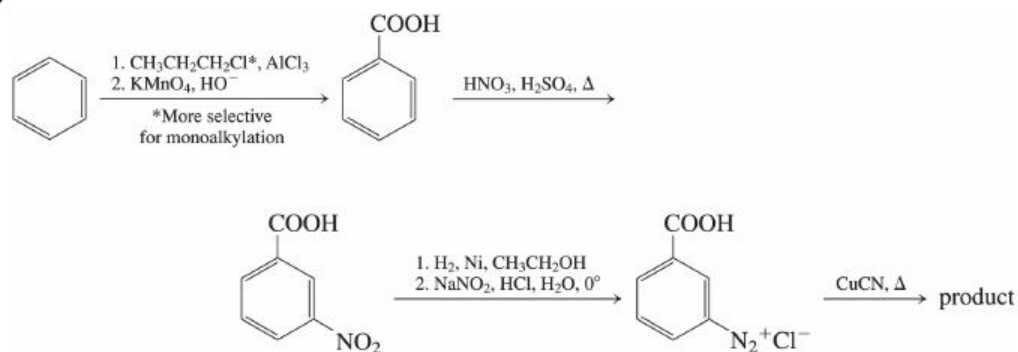


39. Think about benzenediazonium salts as intermediates likely to be useful in these.

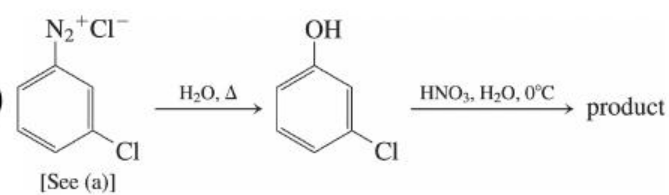
(a)



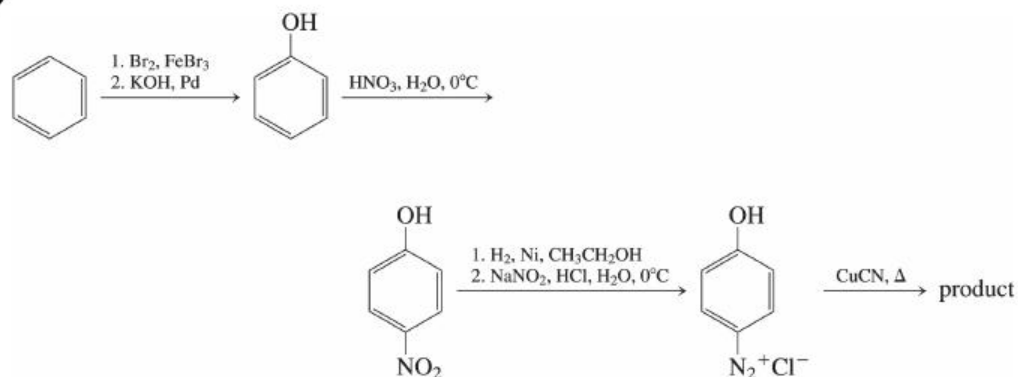
(b)



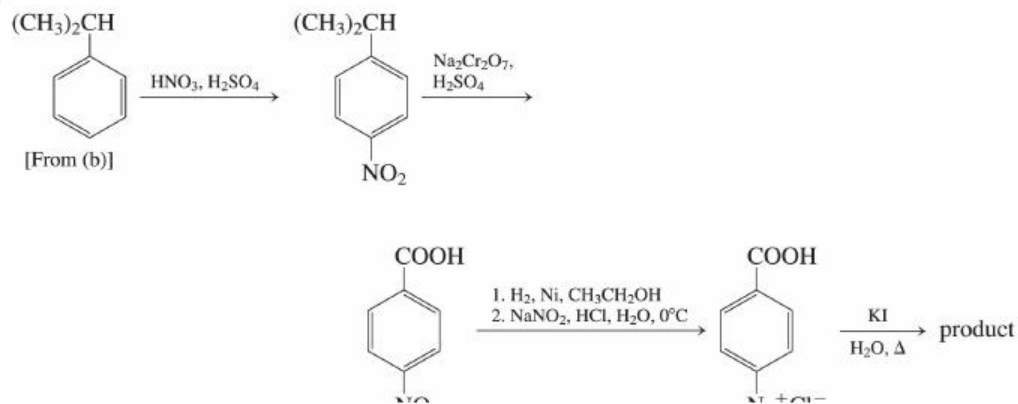
(c)



(d)

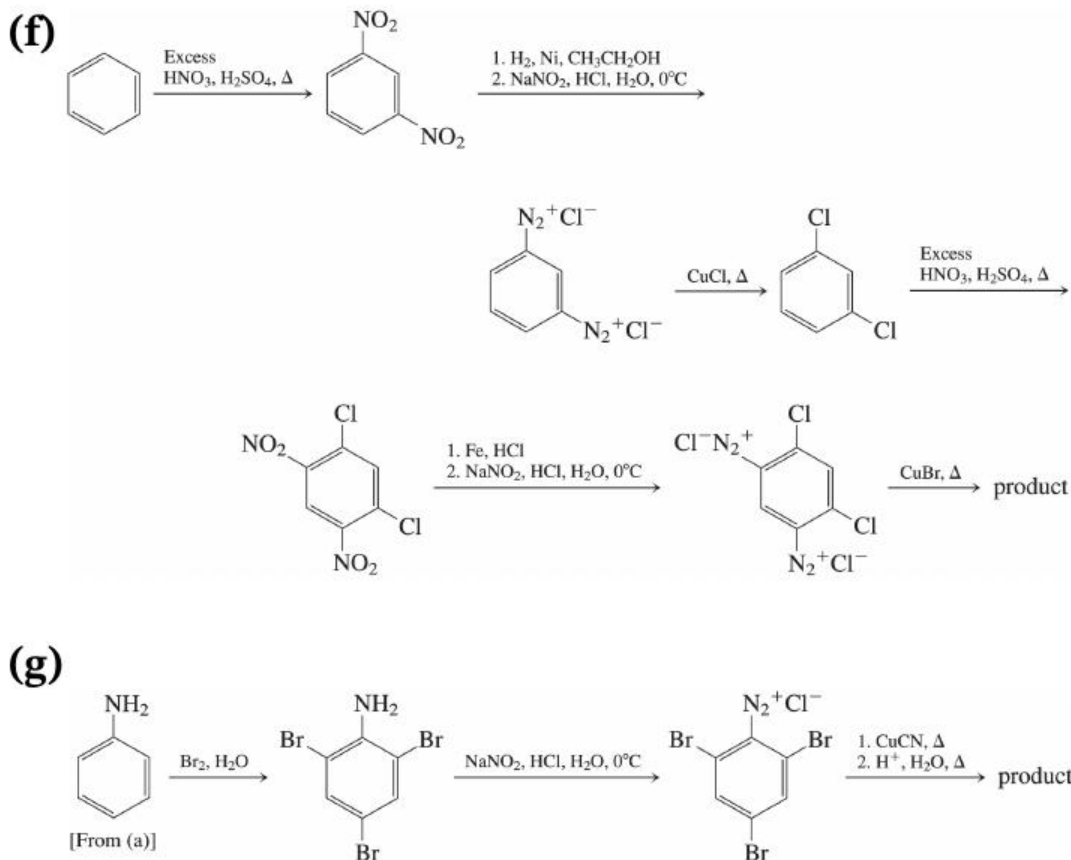


(e)



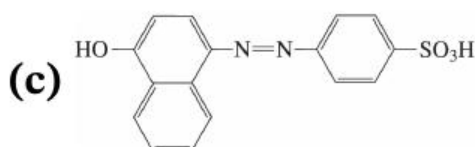
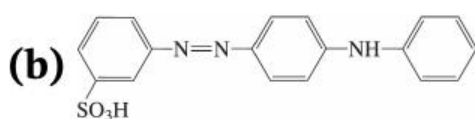


Note concerning the first step: Nitration of methylbenzene gives a lot of ortho product. The larger alkyl group hinders ortho attack and favors para substitution.



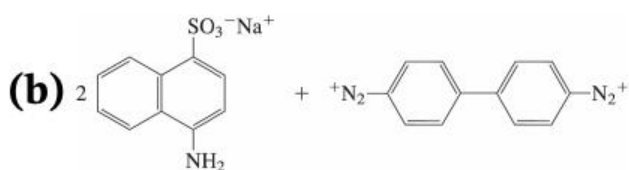
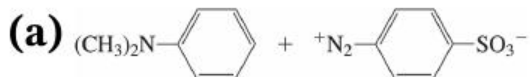
70.

(a)

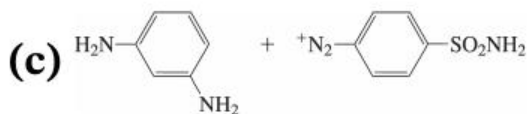


Diazo coupling is generally para to the activating group, if possible.

71.



Note how the benzene ring in the coupling reaction is always strongly activated by OH, NH₂, OH, NH₂, or related groups.

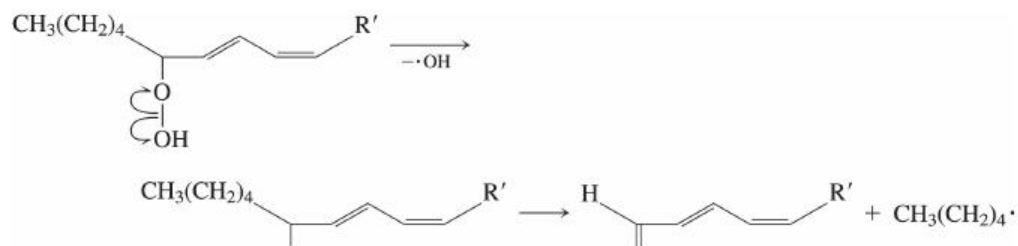


72

(a)

An equation shows a chemical reaction.

(b) Start with the lipid hydroperoxide of linoleic acid shown in [Section 22-9](#) in the reaction labeled “propagation step 2.” Form the alkoxyl radical and finish with β - β -scission.



O·

O

The $\text{CH}_3(\text{CH}_2)_4\cdot$ radical becomes pentane by abstracting a hydrogen atom from any reactive hydrogen donor, such as another lipid molecule.

73. There are several places to start: go step-by-step.

.. Degrees of unsaturation ([Chapter 11](#)).

Urushiol I, $\text{H}_{\text{sat}} = 42 + 2 = 44$; degrees of unsaturation $= (44 - 36) / 2 = 4$
 π bonds or rings

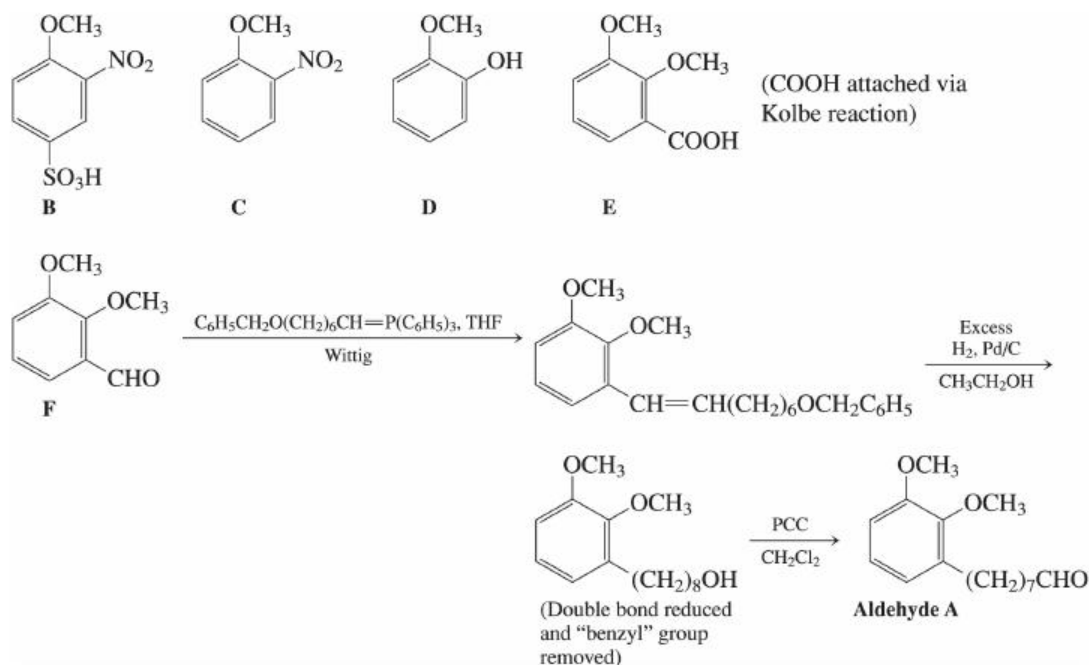
Urushiol II, degrees of unsaturation $= (44 - 34) / 2 = 5$ π bonds or rings

$$\text{degrees of unsaturation} = \frac{(44 - 34)}{2} = 5 \pi \text{ bonds or rings}$$

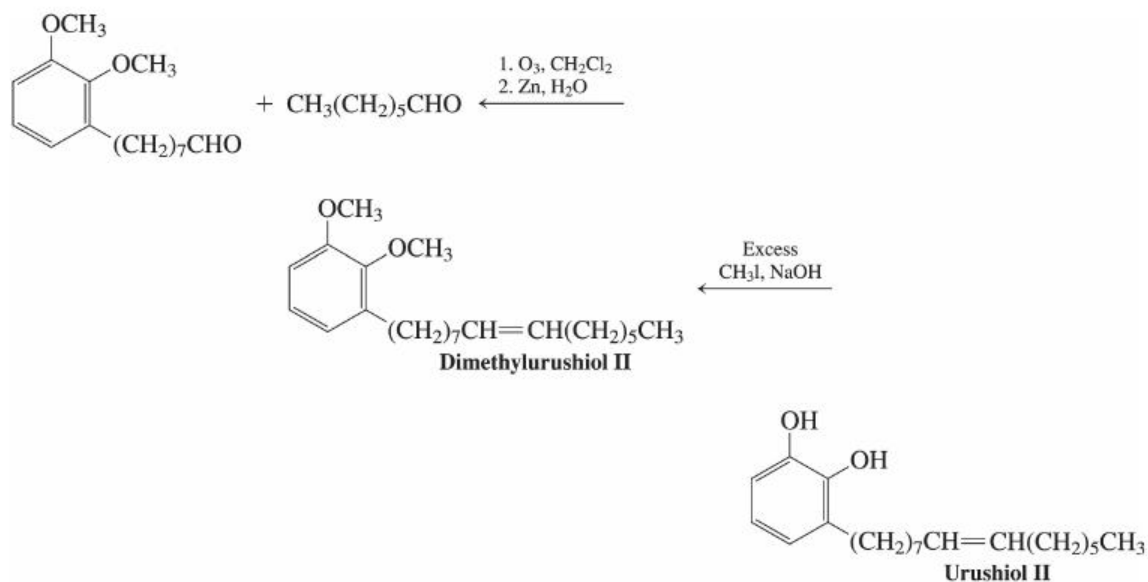
- !. Urushiol II contains only one double bond that is easily hydrogenated.
 The four degrees of unsaturation in urushiol I must be either rings or hard-to-hydrogenate π bonds (like those in a benzene ring).
- !. Urushiol II contains the piece

N_2

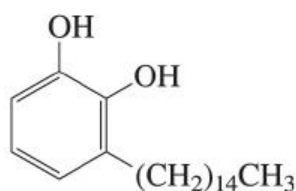
- !. Synthesis of aldehyde A is presented. Here are the structures of the intermediates.



Going back to step 3, you can now work backward to the structure of urushiol II.

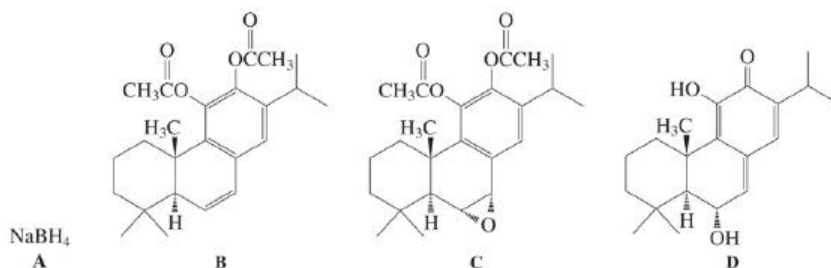


A reconsideration of step 2 shows that urushiol I must be



74. Yes to the first question: A benzylic carbon is oxidized. No to the second, for two reasons. First, the amine is a reactive and oxidizable functional group and would have to be protected. Second, generation of a new stereocenter in exclusively the proper configuration would be difficult, although resolution of a racemic product mixture into its enantiomers would be straightforward.

75.



76. (d) Conversion to phenol takes a *lot* more than boiling in water for two hours.
77. (b)
78. (b)
79. (a)
30. (a) Hydrogen bonding to the nitro group oxygen atom occurs only in the ortho isomer.

23

Ester Enolates and the Claisen Condensation: Synthesis of β - β -Dicarbonyl Compounds; Acyl Anion Equivalents

A large proportion of organic compounds of synthetic and biological importance contain more than one functional group. It is obviously necessary to be able to develop methods for the preparation of such substances. You have already learned quite a bit about the aldol condensation and the systems it forms, β - β -hydroxycarbonyl compounds and conjugated enones ([Chapter 18](#)). By presenting a related reaction of esters, the Claisen condensation, this chapter expands the concept of carbonyl condensation reactions. In addition to being a new method for carbon-carbon bond formation, the Claisen condensation gives rise to a class of compounds, β - β - or 1,3-dicarbonyl compounds, that have their own distinct reactivity and utility. The final chapter section presents a more advanced carbon-carbon bond forming method, showing how certain protected forms of carbonyl carbons may be made nucleophilic and used in displacement and addition reactions.

Outline of the Chapter

23-1 β -Dicarbonyl Compounds: The Claisen Condensation

Construction of the most versatile combination of carbonyl groups.

23-2, 23-3 Synthetic Applications of β -Dicarbonyl Compounds

23-4 Acyl Anion Equivalents: Preparation of α -Hydroxyketones

Reversal of the normal polarity of a functionalized carbon in synthesis.

Keys to the Chapter

23-1. β - β -Dicarbonyl Compounds: The Claisen Condensation

The Claisen condensation is the main method for synthesizing 1,3-dicarbonyl compounds. Analyze this reaction on the basis of its similarities to the aldol condensation ([Section 18-5](#)): It is an enolate + carbonyl process, so bond formation occurs between the α -carbon ~~α -carbon~~ of one carbonyl compound (which may be either an ester or a ketone) and the carbonyl carbon of another (an ester). Note the limitation: Under the conditions given, the reaction works only when the 1,3-dicarbonyl product still possesses a hydrogen on the carbon between the two carbonyl groups. Deprotonation of this acidic H by excess base allows the equilibrium to shift to the product, an excellent example of Le Châtelier's principle in action.

Because there are several types of β - β -carbonyl compounds available from the Claisen condensation, Table 1 presents examples to help you keep them straight.

TABLE 1
Claisen condensations

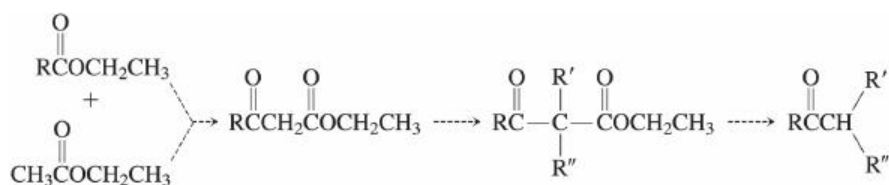
Reaction Partners		Product	
Carbonyl	Enolate	New functional groups	Structure (New bond shown)
$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{CH}_2\text{OCOCCH}_2\text{CH}_3 \\ \uparrow \\ \text{Carbonate} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{COCCH}_2\text{CH}_3 \\ \uparrow \\ \text{General ester} \end{array}$	Ester + ester	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{CH}_3\text{CH}_2\text{OC}-\text{CH}_2\text{COCCH}_2\text{CH}_3 \\ \text{1,3-Diester} \end{array}$
$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{COCCH}_2\text{CH}_3 \\ \uparrow \\ \text{General ester} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{COCCH}_2\text{CH}_3 \\ \uparrow \\ \text{General ester} \end{array}$	Ketone + ester	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{CH}_3\text{C}-\text{CH}_2\text{COCCH}_2\text{CH}_3 \\ \text{3-Ketoester} \end{array}$
$\begin{array}{c} \text{O} \\ \parallel \\ \text{HCOCH}_2\text{CH}_3 \\ \uparrow \\ \text{Formate} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{COCCH}_2\text{CH}_3 \\ \uparrow \\ \text{General ester} \end{array}$	Aldehyde + ester	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{HC}-\text{CH}_2\text{COCCH}_2\text{CH}_3 \\ \text{Formyl ester} \end{array}$
$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{CH}_2\text{OCOCCH}_2\text{CH}_3 \\ \uparrow \\ \text{Carbonate} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{CCH}_3 \\ \uparrow \\ \text{Ketone} \end{array}$	Ester + ketone	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{CH}_3\text{CH}_2\text{OC}-\text{CH}_2\text{CCH}_3 \\ \text{3-Ketoester} \\ \text{(The "hard" way)} \end{array}$
$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{COCCH}_2\text{CH}_3 \\ \uparrow \\ \text{General ester} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{CCH}_3 \\ \uparrow \\ \text{Ketone} \end{array}$	Ketone + ketone	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{CH}_3\text{C}-\text{CH}_2\text{CCH}_3 \\ \text{1,3-Diketone} \end{array}$
$\begin{array}{c} \text{O} \\ \parallel \\ \text{HCOCH}_2\text{CH}_3 \\ \uparrow \\ \text{Formate} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{CCH}_3 \\ \uparrow \\ \text{Ketone} \end{array}$	Aldehyde + ketone	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{HC}-\text{CH}_2\text{CCH}_3 \\ \text{3-Ketoaldehyde} \end{array}$

23-2 and 23-3. Synthetic Applications of β - β -Dicarbonyl Compounds

In Section 23-2 two reactions of β - β -dicarbonyl compounds are presented. The first is alkylation of the readily deprotonated "carbon in the middle." The second pertains to β - β -dicarbonyl compounds where at least one of the carbonyl groups is an ester. Ester hydrolysis leads to a carboxylic acid that readily loses CO_2 (decarboxylation). When this sequence is carried out on a β - β -ketoester, the result is a ketone. When carried out on a diester of propanedioic (malonic) acid, the result is a carboxylic acid. In each case, groups attached in the preliminary alkylation step(s) wind up in the product.

Notice that the acetoacetic ester synthesis makes only **methyl** ketones, because the $\text{CH}_3\text{COCH}_2\text{CO}$ portion of the acetoacetic ester molecule is carried unchanged into the final product. To make other kinds of ketones, other 3-ketoesters must be prepared first. [Problem 47](#) outlines the situation, and the general scheme below illustrates the process.

General 3-Ketoester Synthesis of Ketones



Note that the necessary 3-ketoester comes from a crossed Claisen condensation. As a result, $\text{RCO}_2\text{CH}_2\text{CH}_3\text{RCO}_2\text{CH}_2\text{CH}_3$ must be an ester that does not do Claisen condensations with itself (i.e., R must not have an $\alpha\text{-CH}_2\alpha\text{-CH}_2$ group). Otherwise the crossed condensation will be a disaster.

[Section 23-3](#) reinforces the fact that anions of $\beta\text{-}\beta$ -dicarbonyl compounds are still enolates. Thus, you will find that they do 1,4-additions to $\alpha,\beta\text{-}\alpha,\beta$ -unsaturated carbonyl compounds (Michael additions). Furthermore, these additions can be followed by condensations to complete Robinson annulations, thereby resulting in six-membered rings.

23-4. Acyl Anion Equivalents

How are $\alpha\text{-}\alpha$ -hydroxy ketones made? Because they contain alcohol groups, you might consider methods first introduced in [Chapter 8](#) for the synthesis of alcohols: addition reactions of organometallic (carbanionic) reagents to aldehydes and ketones. However, if you tried to apply this approach, you would encounter a problem: The necessary carbanion

would have the structure $\text{RC} \begin{array}{c} \text{O} \\ || \\ :^- \end{array}$, with a nucleophilic carbonyl carbon.

Such a species is called an acyl anion and is not readily prepared. Indeed, ever since [Chapter 8](#), you have had drilled into your brain the fact that carbonyl carbons are **electrophilic**, not nucleophilic. The whole idea of a nucleophilic acyl anion contradicts everything you've learned.

So, given all that, can anything be done to get around this limitation? Yes: **Carbonyl carbons can be chemically modified to become nucleophilic.** One way is to convert the carbonyl group of an aldehyde into a thioacetal and then deprotonate it with a strong base, forming a 1,3-dithiacyclohexane (1,3-dithiane) anion. Another way is to expose an aldehyde to thiazolium salts. Again, deprotonation of what was originally a carbonyl carbon may follow, giving a nucleophilic anion. Either way, you have reversed the normal polarity of this carbon atom. Once such an anion (an acyl anion equivalent) has been formed, it can add to the normal, electrophilic carbonyl group of another molecule in the usual way. **The result is a new carbon-carbon bond between carbons that both started out with the same polarity (electrophilic).** This is **important.**

Application of polarity reversal in organic chemistry isn't magical. All you've done is turn carbonyl groups into new functions that are able to support negative charges. Then, after finishing with them, you've changed them back to carbonyl groups again. Still, this material can be troublesome to learn. You might try the following for study purposes: Write down several aldehydes and their (hard to make) acyl anions. Next, follow along one of the two general reaction sequences shown in the text section, drawing the corresponding acyl anion equivalent, adding it to another carbonyl group, and regenerating the original carbonyl carbon. Notice in particular how the original carbonyl carbon has been modified so as to be able to support a negative charge and become a nucleophile.

The practice will be good for you.